

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November 2013

EC 203 Solid State Electronics

Date: 26.11.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

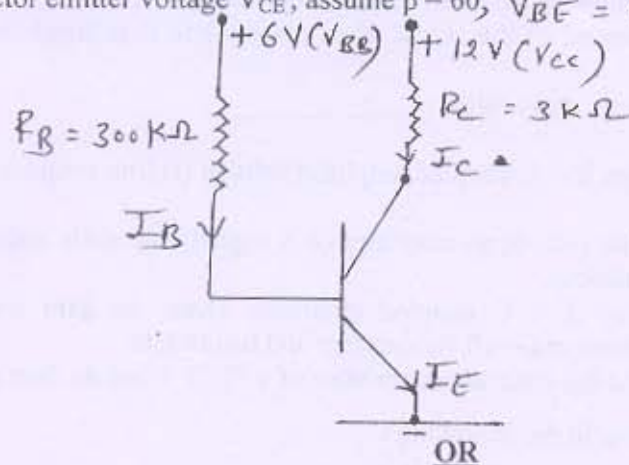
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Do as Direct.

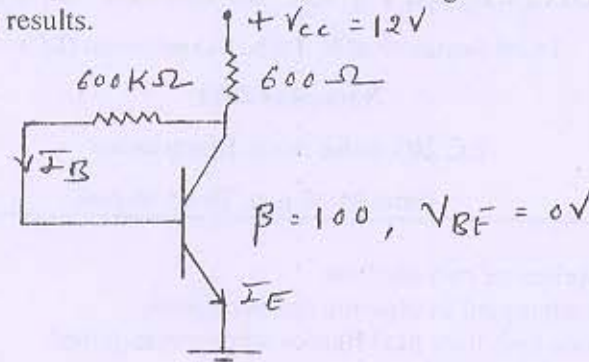
- (a) Define α and β for a transistor and establish the relationship between them. [04]
- (b) Define Static and Dynamic resistance of the diode with the help of neat sketch. [02]
- (c) Define energy band. [01]

- Q - 2 (a) Describe the important properties of pure semiconductors. [03]
- (b) What do you understand by hole in a semiconductor? Define hole current. [03]
- (c) If V_m is the peak values of voltage across the secondary in a half wave rectifier (HWR), what is the value of d.c. output derive this relation. [02]
- (d) What do you understand by an 'ideal diode'? Draw its $V - I$ characteristics. Explain its switching action. [02]
- (e) A base resistor biasing circuit is shown in Fig. Determine the collector current I_C and collector emitter voltage V_{CE} , assume $\beta = 60$, $V_{BE} = 0.7V$. [04]



- Q - 2 (a) On the basis of conductivity, classify the solid; Explain their behavior on the basis of energy band phenomenon. [03]
- (b) Discuss the behavior of a PN junction under reverse biased. [03]
- (c) Why a bridge type full wave rectifier is preferred over a centre tap full wave rectifier. [02]
- (d) What do you mean by ripples in relation to the rectifier output? [02]

- (e) Determine the emitter current and collector voltage in the circuit shown in Fig. [04]
comment on the results.



- Q - 3 (a) Draw common emitter hybrid - pi model at low frequency and derive the [07]
transconductance equation.
(b) What will be the variation in following parameter? When variation with increasing, [07]
collector current (I_C), collector to emitter voltage (V_{CE}) and temperature (T).

g_m , $r_{bb'}$, $r_{b'e}$, C_e , C_c , h_{fe} , h_{ie}
OR

- Q - 3 (a) Write a note on hybrid - pi capacitance. [07]
(b) Write a note on hybrid - pi conductance. [07]

SECTION - II

Q - 4 Do as Direct

- (a) A bridge rectifier circuit is not suitable for _____ voltage amplification. [01]
(b) "In multistage amplifier, the input impedance of an amplifier stage should very high [02]
and output impedance must be very low" Justify this statement.
(c) Explain how phase reversal of the signal takes place when it is amplified by single [03]
stage amplifier.
(d) In h - parameters analysis, h_{fe} is called _____. [01]

- Q - 5 (a) Why does the gain of an R - C coupled amplifier falls in (i) low frequency range (ii) [02]
high frequency range.
(b) We prefer to express the gain of an amplifier on a logarithmic scale rather than on a [02]
linear scale. State the reasons.
(c) Draw the circuit of an R - C coupled amplifier. Draw its gain v/s frequency [04]
characteristics and indicate cut - off frequencies and bandwidth.
(d) How will you determine the drain characteristics of a FET? What do they indicate? [04]
(e) Select the correct answer in the following : [02]

(i) A FET is characterized by (Option : Current gain, voltage gain, power gain,
none of these)

(ii) Input impedance of MOSFET is (option : less than that of FET but more than
BJT, more than that of FET and BJT, more than that of FET but less than
BJT, less than that of FET and BJT)

OR

- Q – 5 (a) Explain what will happen to the voltage gain of an amplifier if the bypass capacitor C_E is opened. [02]
- (b) Show the various currents and voltages in a single stage transistor amplifier and relate them. [02]
- (c) Explain briefly the following distortions. [04]
- (i) Frequency distortion
 - (ii) Phase distortion
- (d) Draw JFET low frequency small signal model and derive the expression for transconductance. [04]
- (e) Explain the working principle of MOSFET. [02]
- Q – 6 (a) Determine the transconductance of a FET if its amplification factor is 96 and drain resistance is 32 K ohms. [04]
- (b) 1. Draw the hybrid model for three different configuration of bipolar junction transistor. [05]
2. Define operating point of transistor. What do you understand by stabilization of operating point? Mention the essentials of biasing. [05]

OR

- Q – 6 (a) Determine the value of transconductance of a FET, when the drain current changes from 1 mA to 1.9 mA with a change in gate – source voltage from -3.3 V to -3V. [04]
- (b) Draw the generalized hybrid model for BJT and derive its following parameter. [10]
- 1. current gain
 - 2. input impedance
 - 3. voltage gain
 - 4. output impedance
